

LAB - DB Schema University Management

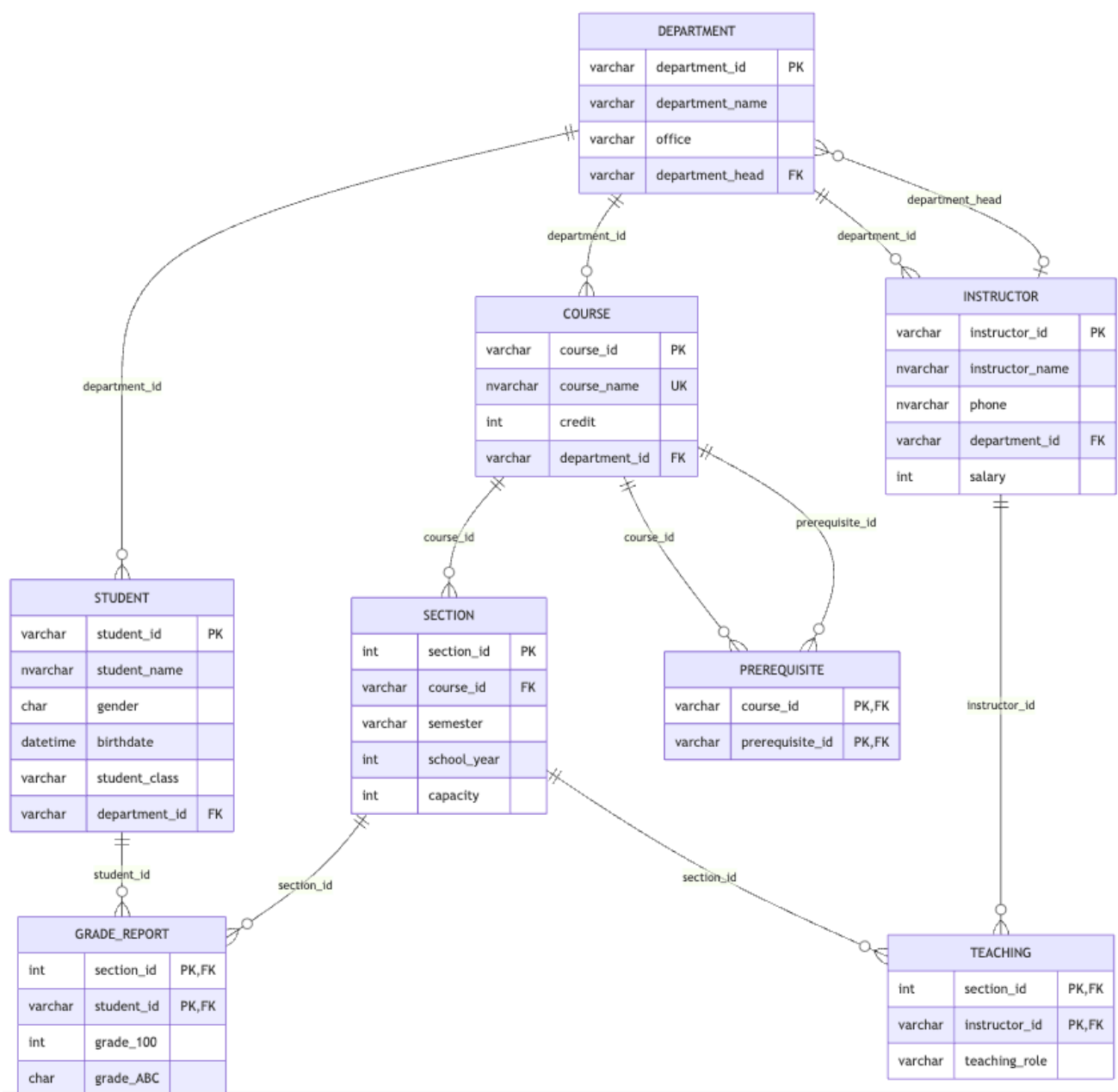


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The University Management Database Specification

(Extended from The University Database in Fundamentals of Database Systems)

1. Database schema



Instructor

- **instructor_id: varchar(9), primary key**
- instructor_name: nvarchar(50), not null
- phone: nvarchar(9), not null
- department_id: varchar(5), reference to department_id of the Department table
- salary: int (greater than 0)

Department

- **department_id: varchar(5), primary key**
- department_name: nvarchar(50), not null
- office: varchar(5)
- department_head: varchar(9), reference to instructor_id of the Instructor table

Student

- **student_id: varchar(9), primary key**
- student_name: nvarchar(50), not null
- gender: char(1) (F, M, O)
- birthdate: datetime
- class: varchar(5)
- department_id: varchar(5), reference to ID of the Department table

Course

- **course_id: varchar(9), primary key**
- course_name: nvarchar(50), not null, unique
- credit: int (greater than 0)
- department_id: varchar(5), reference to department_id of the Department table

Section

- **section_id: int, not null, primary key**
- course_id: varchar(9), not null, reference to course_id of the Course table
- semester: varchar(9), not null
- year: int, not null
- capacity: int (greater than 0)
- (course_id, semester, year) combination is unique

Teaching

- section_id: int, not null, reference to section_id of the Section table
- instructor_id: varchar(9), not null, reference to instructor_id of the Instructor table
- role: varchar(9) (Lecturer, TA)
- **(section_id, instructor_id) combination is primary key**

GradeReport

- section_id: int, not null, reference to section_id of the Section table

- student_id: varchar(9), not null, reference to student_id of the Student table
- grade_100: int (greater or equal to 0)
- grade_ABC: char(1) (A, B, C, D, E, or F)
- **(section_id, student_id) combination is primary key**

Prerequisite

- course_id: varchar(9), not null, reference to course_id of the Course table
- prerequisite_id: varchar(9), not null, reference to course_id of the Course table
- **(course_id, prerequisite_id) is primary key**

2. Data Sample

DEPARTMENT

department_id	department_name	office	department_head
AI	Artificial Intelligence	I86	I009
CS	Computer Science	I81	I001
IS	Information System	I84	I004
NW	Network	I87	NULL
SE	Software Engineering	I82	I003

STUDENT

student_id	student_name	gender	birthdate	class	department_id
ST001	Nguyen Ai Linh	F	2002-12-01 00:00:00.000	NULL	CS
ST002	Tran Thanh Sang	M	2003-09-02 00:00:00.000	NULL	CS
ST003	Huynh Thanh Phong	M	2001-05-03 00:00:00.000	NULL	SE
ST004	Hoang Nhat Linh	F	2002-05-10 00:00:00.000	NULL	SE
ST005	Le Ba Khanh	M	2001-11-12 00:00:00.000	NULL	SE
ST006	Ly Quoc Phong	M	2000-08-12 00:00:00.000	NULL	SE
ST007	Tran Thanh An	F	2000-11-09 00:00:00.000	NULL	IS
ST008	Le Nha Thu	F	2002-08-09 00:00:00.000	NULL	IS
ST009	Ho Ngoc Anh	F	2003-11-01 00:00:00.000	NULL	AI
ST010	Nguyen Thanh Son	M	2003-12-05 00:00:00.000	NULL	NW

INSTRUCTOR

instructor_id	instructor_name	phone	department_id	salary
I001	Dang Huynh Bao Khanh	080913213	CS	1000
I002	Alex Grant	082412613	CS	2000
I003	Tran Hoang Lan	080921234	SE	1500
I004	Nguyen Ngoc Khanh	090245613	IS	1500
I005	James Cobb	092193213	SE	2000
I006	Le Khanh	090799131	IS	2200
I007	Vu Ngoc Bao	090511342	SE	2100
I008	Tran Hong An	099912353	NW	1900
I009	Nguyen Hai Lam	080911234	AI	1500
I010	Dang Hoang Phong	090233451	AI	2300

COURSE

course_id	course_name	credit	department_id
CS01	Databases	4	IS
CS02	Database Management System	4	IS
CS03	Introduction to Programming	4	SE
CS04	Object-Oriented Programming	4	SE
CS05	Basic Network	4	NW
CS06	Advanced Network	4	NW
CS07	Introduction to Artificial Intelligence	4	AI
CS08	Introduction to Machine Learning	4	CS
CS09	Computer Vision	4	CS
CS10	Robotics	4	AI

PREREQUISITE

course_id	prerequisite_id
CS02	CS01
CS04	CS03
CS06	CS05
CS08	CS07
CS09	CS07
CS10	CS07

SECTION

section_id	course_id	semester	school_year	capacity
1	CS01	Fall	2022	30
2	CS02	Fall	2022	30
3	CS03	Fall	2022	30
4	CS04	Fall	2022	30
5	CS01	Spring	2022	20
6	CS02	Spring	2022	20
7	CS03	Spring	2022	20
8	CS05	Spring	2022	20
9	CS05	Fall	2023	12
10	CS06	Fall	2023	12
11	CS07	Fall	2023	12

TEACHING

section_id	instructor_id	teaching_role
1	I004	Lecturer
1	I006	TA
2	I004	Lecturer
2	I007	TA
3	I003	TA
3	I005	Lecturer
4	I005	Lecturer
4	I009	TA
5	I004	Lecturer
5	I006	TA
6	I004	Lecturer
7	I005	Lecturer
8	I008	Lecturer

GRADEREPORT

section_id	student_id	grade_100	grade_ABC
1	ST001	80	B
1	ST002	82	B
1	ST003	35	F
1	ST004	60	F
1	ST005	100	A
1	ST006	100	A
1	ST007	90	A
1	ST008	52	F
1	ST009	36	F
1	ST010	99	A
2	ST001	77	C
2	ST002	84	B
2	ST003	60	F
2	ST004	53	F
2	ST005	99	A
2	ST006	93	A
2	ST007	82	B
2	ST008	63	F
2	ST009	62	F
2	ST010	88	B